

Class18

Leah Kim A16973745

Table of contents

Background	1
Enter the CMI-PB Project	3
Focus on IgG level	10

Background

Pertussis (aka whooping cough) is a common lung infection caused by the bacteria *B. Pertussis*.

The CDC tracs cases of Pertussis in the US: <https://www.cdc.gov/pertussis/php/surveillance/pertussis-cases-by-year.html>

##Examininc cases of Pertussis by year

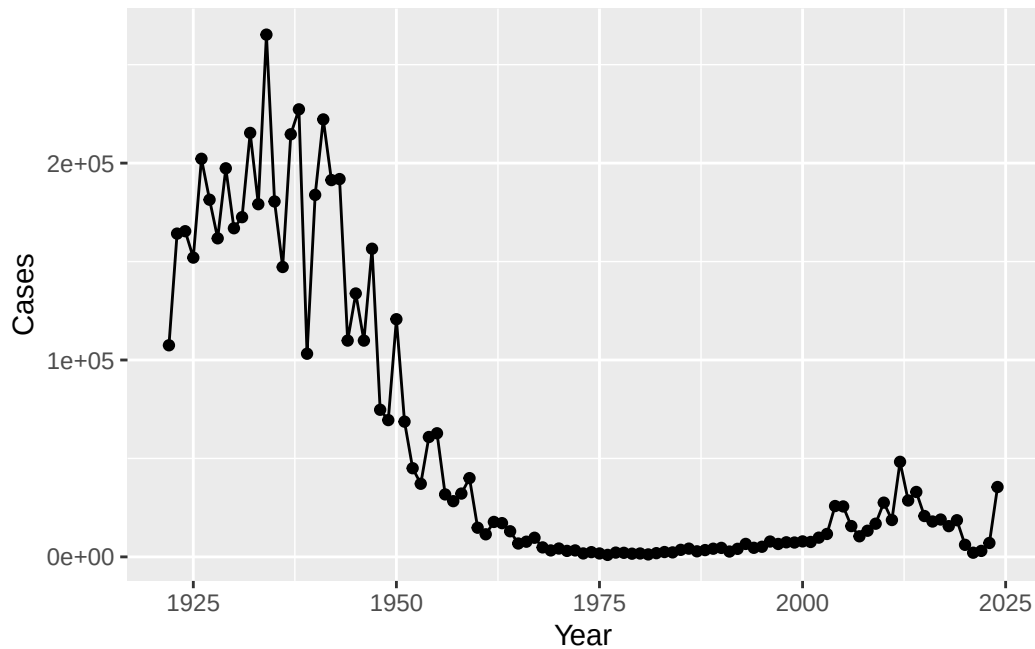
We can use the **datapasta** package to scrape case numbers from the CDC website

. Q Make a plot of pertussis cases per year using ggplot

```
library(ggplot2)

cases <- ggplot(cdc) +
  aes(x = Year, y = Cases) +
  geom_point() +
  geom_line()

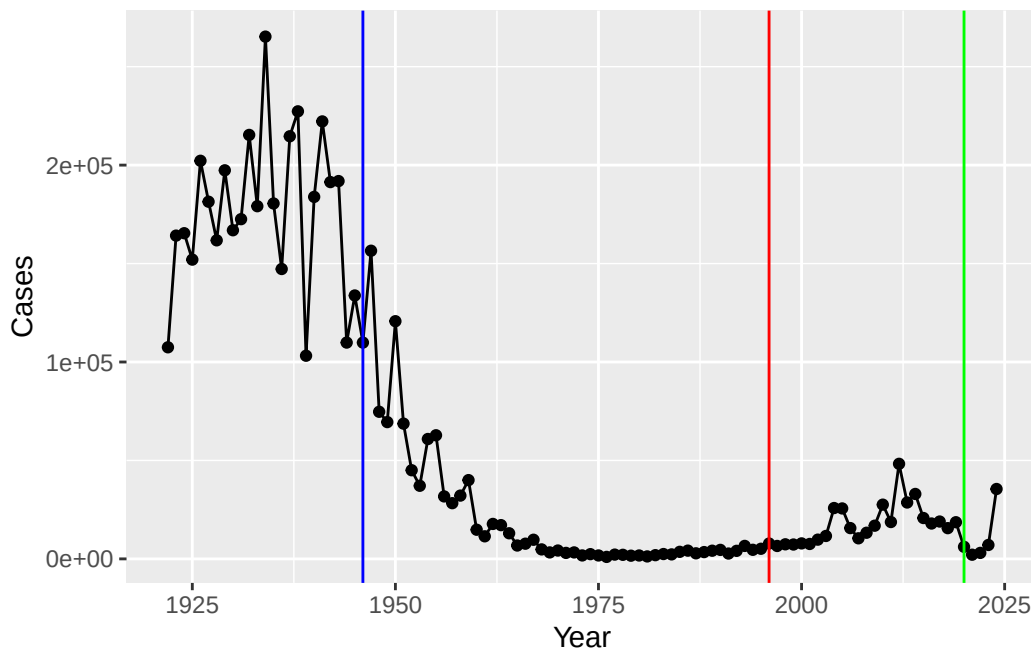
cases
```



Q2. Add some key time points in our history of interaction with Pertussis. These include wP rollout (the first vaccine) in 1946 and the switch to aP in 1996, and the COVID lockdown in 2020.

We can use `geom_vline()` for this.

```
cases +
  geom_vline(xintercept = 1946, col = "blue") +
  geom_vline(xintercept = 1996, col = "red") +
  geom_vline(xintercept = 2020, col = "green")
```



Mounting evidence suggests that the newer **aP** vaccine is less effective over the loong term than the older **wP** vaccine that it replaced. In other words, vaccine proection wanes more rapidly with aP or with wP.

Enter the CMI-PB Project

The goal of the CMI-PB (Computational Models of Immunity - Pertussis Boost) project is to investigate how the immune responds differently to aP vs wP vaccinated individuals and be able to predict this at an early stage.

CMI-PB makes all their collected data freely available and they store it in a databse composed of different tables. Here, we will access a few of these.

We can use the **jsonlite** package to read this data.

```
library(jsonlite)

subjects <- read_json("https://www.cmi-pb.org/api/v5_1/subject", simplifyVector = TRUE)

head(subjects)
```

	subject_id	infancy_vac	biological_sex	ethnicity	race
1	1	wP	Female	Not Hispanic or Latino	White

2	2	wP	Female Not Hispanic or Latino White
3	3	wP	Female Unknown White
4	4	wP	Male Not Hispanic or Latino Asian
5	5	wP	Male Not Hispanic or Latino Asian
6	6	wP	Female Not Hispanic or Latino White

	year_of_birth	date_of_boost	dataset
1	1986-01-01	2016-09-12	2020_dataset
2	1968-01-01	2019-01-28	2020_dataset
3	1983-01-01	2016-10-10	2020_dataset
4	1988-01-01	2016-08-29	2020_dataset
5	1991-01-01	2016-08-29	2020_dataset
6	1988-01-01	2016-10-10	2020_dataset

Q. How many subjects are there?

```
nrow(subjects)
```

```
[1] 172
```

Q. How many aP and wP subjects are there?

```
table(subjects$infancy_vac)
```

```
aP wP
87 85
```

Q. How many males and females are there in the dataset?

```
table(subjects$biological_sex)
```

```
Female  Male
112     60
```

Q. How about gender and race?

```
table(subjects$race, subjects$biological_sex)
```

	Female	Male
American Indian/Alaska Native	0	1
Asian	32	12
Black or African American	2	3
More Than One Race	15	4
Native Hawaiian or Other Pacific Islander	1	1
Unknown or Not Reported	14	7
White	48	32

Q. Is this representative of the US population?

No. This is more the demographic of UCSD students (since this data was gathered from UCSD students).

Let' read another database table from CMI-PB.

```
specimens <- read_json("https://www.cmi-pb.org/api/v5_1/specimen", simplifyVector = TRUE)
ab_data <- read_json("http://cmi-pb.org/api/v5_1/plasma_ab_titer", simplifyVector = TRUE)
```

Wee peek at these:

```
head(specimens)
```

```
specimen_id subject_id actual_day_relative_to_boost
1           1           1                        -3
2           2           1                         1
3           3           1                         3
4           4           1                         7
5           5           1                        11
6           6           1                        32
planned_day_relative_to_boost specimen_type visit
1                           0         Blood      1
2                           1         Blood      2
3                           3         Blood      3
4                           7         Blood      4
5                          14         Blood      5
6                          30         Blood      6
```

```
head(ab_data)
```

	specimen_id	isotype	is_antigen_specific	antigen	MFI	MFI_normalised
1	1	IgE	FALSE	Total	1110.21154	2.493425
2	1	IgE	FALSE	Total	2708.91616	2.493425
3	1	IgG	TRUE	PT	68.56614	3.736992
4	1	IgG	TRUE	PRN	332.12718	2.602350
5	1	IgG	TRUE	FHA	1887.12263	34.050956
6	1	IgE	TRUE	ACT	0.10000	1.000000

	unit	lower_limit_of_detection
1	UG/ML	2.096133
2	IU/ML	29.170000
3	IU/ML	0.530000
4	IU/ML	6.205949
5	IU/ML	4.679535
6	IU/ML	2.816431

We want to join these tables to get all our information together. For this, we will use the **dplyr** package and the `inner_join()` function.

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

`filter`, `lag`

The following objects are masked from 'package:base':

`intersect`, `setdiff`, `setequal`, `union`

```
meta <- inner_join(subjects, specimens)
```

Joining with `by = join\_by(subject\_id)`

```
head(meta)
```

	subject_id	infancy_vac	biological_sex	ethnicity	race
1	1	wP	Female	Not Hispanic or Latino	White
2	1	wP	Female	Not Hispanic or Latino	White
3	1	wP	Female	Not Hispanic or Latino	White
4	1	wP	Female	Not Hispanic or Latino	White
5	1	wP	Female	Not Hispanic or Latino	White
6	1	wP	Female	Not Hispanic or Latino	White

	year_of_birth	date_of_boost	dataset	specimen_id
1	1986-01-01	2016-09-12	2020_dataset	1
2	1986-01-01	2016-09-12	2020_dataset	2
3	1986-01-01	2016-09-12	2020_dataset	3
4	1986-01-01	2016-09-12	2020_dataset	4
5	1986-01-01	2016-09-12	2020_dataset	5
6	1986-01-01	2016-09-12	2020_dataset	6

	actual_day_relative_to_boost	planned_day_relative_to_boost	specimen_type
1	-3	0	Blood
2	1	1	Blood
3	3	3	Blood
4	7	7	Blood
5	11	14	Blood
6	32	30	Blood

	visit
1	1
2	2
3	3
4	4
5	5
6	6

One more join to get ab_data and mta all together.

```
abdata <- inner_join(ab_data, meta)
```

Joining with `by = join\_by(specimen\_id)`

```
head(abdata)
```

	specimen_id	isotype	is_antigen_specific	antigen	MFI	MFI_normalised
1	1	IgE	FALSE	Total	1110.21154	2.493425
2	1	IgE	FALSE	Total	2708.91616	2.493425
3	1	IgG	TRUE	PT	68.56614	3.736992

4	1	IgG	TRUE	PRN	332.12718	2.602350
5	1	IgG	TRUE	FHA	1887.12263	34.050956
6	1	IgE	TRUE	ACT	0.10000	1.000000

	unit	lower_limit_of_detection	subject_id	infancy_vac	biological_sex
1	UG/ML	2.096133	1	wP	Female
2	IU/ML	29.170000	1	wP	Female
3	IU/ML	0.530000	1	wP	Female
4	IU/ML	6.205949	1	wP	Female
5	IU/ML	4.679535	1	wP	Female
6	IU/ML	2.816431	1	wP	Female

	ethnicity	race	year_of_birth	date_of_boost	dataset
1	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
2	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
3	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
4	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
5	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
6	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset

	actual_day_relative_to_boost	planned_day_relative_to_boost	specimen_type
1	-3		Blood
2	-3		Blood
3	-3		Blood
4	-3		Blood
5	-3		Blood
6	-3		Blood

	visit
1	1
2	1
3	1
4	1
5	1
6	1

```
dim(abdata)
```

```
[1] 61956    20
```

Q. How many antibody isotypes are there in the dataset?

```
table(abdata$isotype)
```

```

IgE  IgG  IgG1  IgG2  IgG3  IgG4
6698 7265 11993 12000 12000 12000

```


Q. How many different antigens are measured in the dataset?

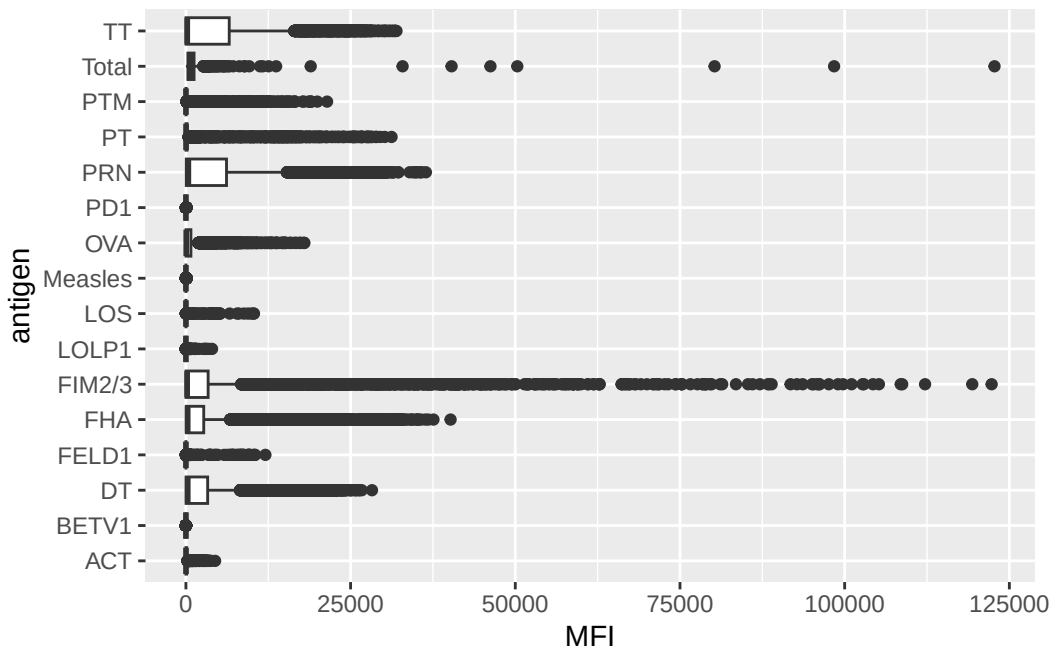
```
table(abdata$antigen)
```

ACT	BETV1	DT	FELD1	FHA	FIM2/3	LOLP1	LOS	Measles	OVA
1970	1970	6318	1970	6712	6318	1970	1970	1970	6318
PD1	PRN	PT	PTM	Total	TT				
1970	6712	6712	1970	788	6318				

Q. Make a boxplot of antigen levels across the whole dataset.

```
ggplot(abdata) +  
  aes(x = MFI, y = antigen) +  
  geom_boxplot()
```

Warning: Removed 1 row containing non-finite outside the scale range (`stat\_boxplot()`).

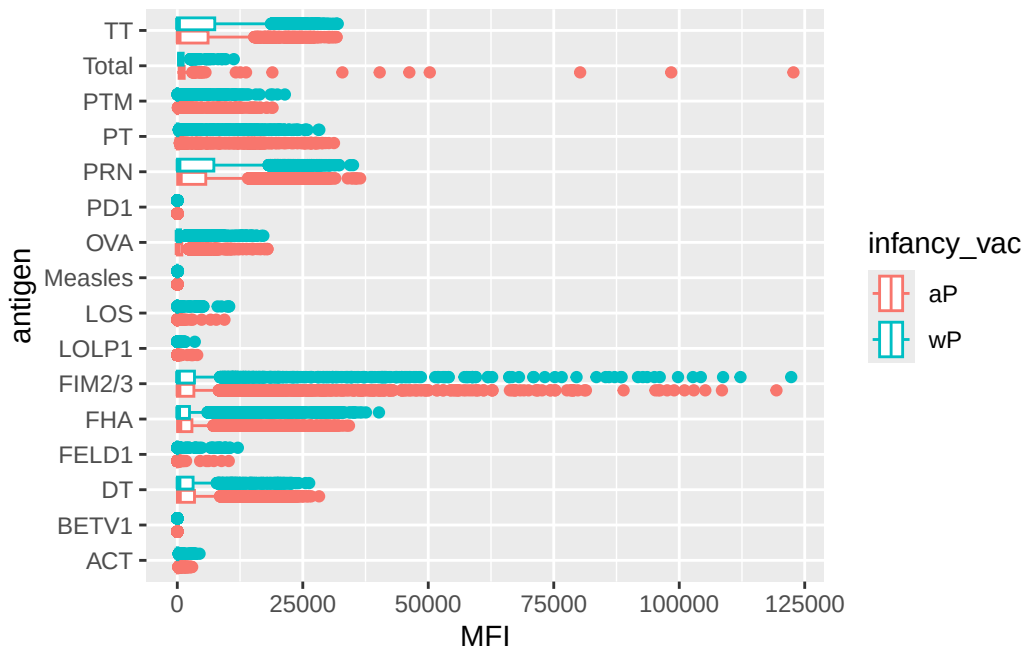


Measles is flat as it is meant to be a kind of control. Meanwhile, antigens such as TT and ACT are present since the vaccine contains them.

Q. Are there obvious differences between aP and dP values?

```
ggplot(abdata) +  
  aes(x = MFI, y = antigen, col = infancy_vac) +  
  geom_boxplot()
```

Warning: Removed 1 row containing non-finite outside the scale range
(`stat\_boxplot()`).



There are some differences but nothing that significant.

Focus on IgG level

IgG is the most abundant antibody in blood. With four sub-classes (IgG1 to IgG4) crucial for long term-immunity and responding to bacterial and viral infections.

```
igg <- abdata |> filter(isotype == "IgG")  
head(igg)
```

	specimen_id	isotype	is_antigen_specific	antigen	MFI	MFI_normalised
1	1	IgG	TRUE	PT	68.56614	3.736992
2	1	IgG	TRUE	PRN	332.12718	2.602350
3	1	IgG	TRUE	FHA	1887.12263	34.050956
4	19	IgG	TRUE	PT	20.11607	1.096366
5	19	IgG	TRUE	PRN	976.67419	7.652635
6	19	IgG	TRUE	FHA	60.76626	1.096457

	unit	lower_limit_of_detection	subject_id	infancy_vac	biological_sex
1	IU/ML	0.530000	1	wP	Female
2	IU/ML	6.205949	1	wP	Female
3	IU/ML	4.679535	1	wP	Female
4	IU/ML	0.530000	3	wP	Female
5	IU/ML	6.205949	3	wP	Female
6	IU/ML	4.679535	3	wP	Female

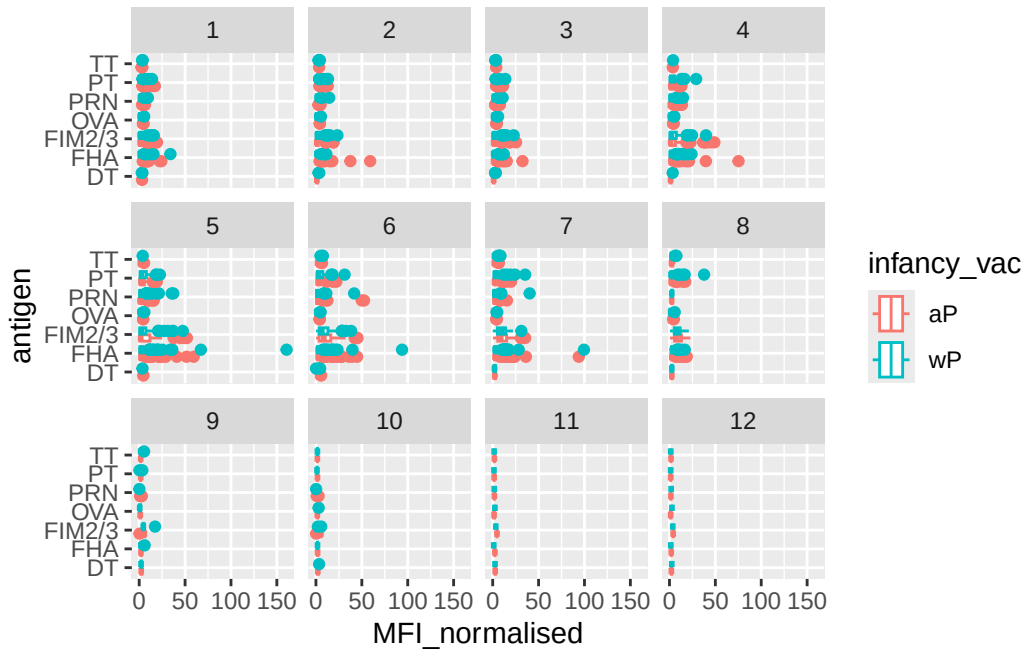
	ethnicity	race	year_of_birth	date_of_boost	dataset
1	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
2	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
3	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
4	Unknown	White	1983-01-01	2016-10-10	2020_dataset
5	Unknown	White	1983-01-01	2016-10-10	2020_dataset
6	Unknown	White	1983-01-01	2016-10-10	2020_dataset

	actual_day_relative_to_boost	planned_day_relative_to_boost	specimen_type
1	-3		Blood
2	-3		Blood
3	-3		Blood
4	-3		Blood
5	-3		Blood
6	-3		Blood

	visit
1	1
2	1
3	1
4	1
5	1
6	1

Same boxplot of anitgens as before.

```
ggplot(igg) +
  aes(x = MFI_normalised, y = antigen, col = infancy_vac) +
  geom_boxplot() + facet_wrap(~visit)
```



Focus in further on just one of these antigens - Let's pick PT (Pertussis toxin, one of the main toxins of the bacteria).

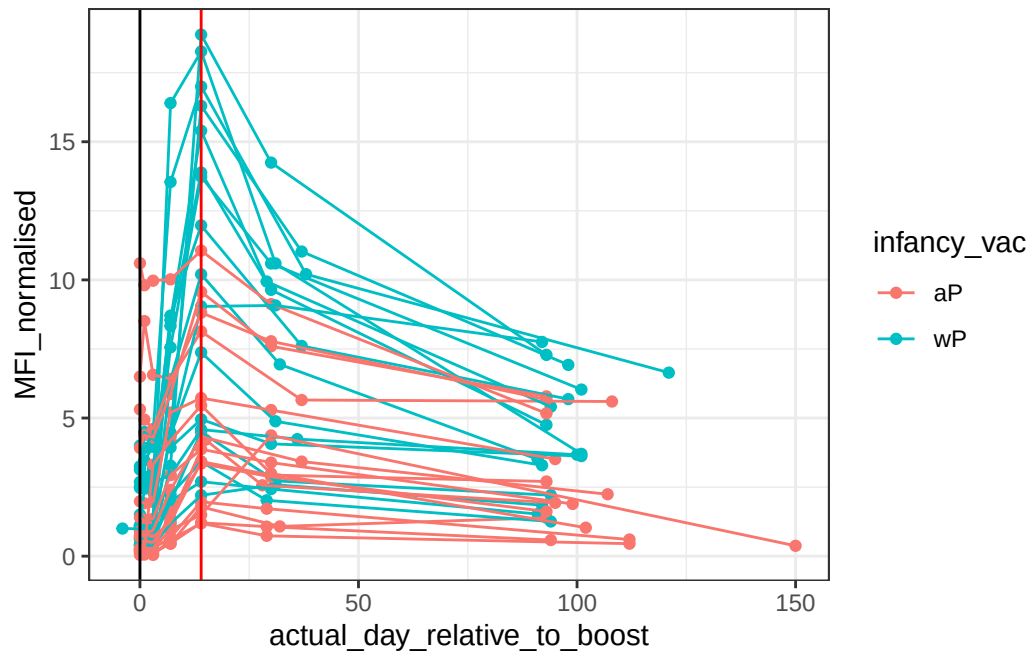
```
table(igg$dataset)
```

```
2020_dataset 2021_dataset 2022_dataset 2023_dataset
      1182      1617      1456      3010
```

```
pt_igg <- abdata |> filter(isotype == "IgG", antigen == "PT", dataset == "2021_dataset")
dim(pt_igg)
```

```
[1] 231  20
```

```
ggplot(pt_igg) + aes(x = actual_day_relative_to_boost, y = MFI_normalised, col = infancy_vac)
  geom_point() +
  geom_line() +
  theme_bw() +
  geom_vline(xintercept = 0) +
  geom_vline(xintercept = 14, col = "red")
```



On day 14, there is a visible peak in the MFI, and the paths show that generally, wP individuals have a higher MFI normalized value than aP.